

Milestone 3 – Cache Manager, Due October 14, 2026 4:00 PM Pacific

The following files will be provided to you, for completion of your milestone:

- `cache_manager.h` // header file containing cache manager class
- `dll_node.cpp` // dll node file constructor
- `dll_node.h` // header file containing doubly linked list structure
- `doubly_linked_list.h` // header file containing doubly linked list class
- `hash_node.cpp` // hash node file constructor
- `hash_node.h` // header file containing hash node structure
- `hash_table.h` // header file containing hash table class
- `json.hpp` // header file for processing json files
- `outputFile – testcase 1.txt` // expected output for testcase 1
- `milestone3.h` // header file for milestone3 method declarations
- `milestone3.json` // json file containing test cases and its transactions
- `milestone3_config.json` // json configuration (properties) file
- `milestone3.cpp` /* cpp file containing main, which does the following:

- Reads configuration file (json format) to:
 - retrieve `inputFile` (test case file (json format))
 - retrieve `outputFile` (text file containing generated output)
 - retrieve `errorLogFile` (text file containing error messages)
- process `inputFile` test cases
- write output to `outputFile` */

1. Write a Unified Modeling Language (UML) Class Diagram and Sequence Diagram for this milestone.
2. Write a FIFO list and basic hash table implementation, which uses the files listed above, and includes the following in separate cpp files:

- `cache_manager.cpp` – implementation file that contains the following methods:
 1. `getTable` - return the hash table
 2. `getList` - return the FIFO list
 3. `getSize` - return the number of items in the `CacheManager`
 4. `isEmpty` - Check if the `HashTable` is empty
 5. `add` - adds a new node to the `CacheManager`; if number of items $\geq$ `FIFOListSize`, then remove the oldest entry
 6. `remove` – remove node with key value
 7. `clear` - remove all entries from the `CacheManager`
 8. `getItem` - retrieve item from the `CacheManager`
 9. `getMaxCacheSize` - retrieve max size of cache from the `CacheManager`

- 10. contains - check if a node with key exists in the table
- 11. printCache - prints the current contents of the cache to console and output file
- doubly_linked_list.cpp – implementation file that contains the following methods:
 - 12. getSize - Get the size of the list
 - 13. isEmpty - Check if the list is empty
 - 14. insertAtHead - Adds a new node at the beginning of the list
 - 15. insertAtTail - Adds a new node at the end of the list
 - 16. remove - Searches for a node with a specific value and deletes it from the list
 - 17. removeHeaderNode - Removes header node
 - 18. removeTailNode - Removes tail node
 - 19. moveNodeToHead - Moves a specific node to the front
 - 20. moveNodeToTail - Moves a specific node to the end
 - 21. clear - Clear the list (delete all nodes)
 - 22. printList - prints the list from head to tail to the console and output file
 - 23. reversePrintList - prints the list from tail to head to the console and output file
- hash_table.cpp – implementation file that contains the following methods:
 - 24. getTable - return the hash table
 - 25. getSize - return the size of the hash table
 - 26. calculateHashCode - perform hashing function
 - 27. isEmpty - Check if the HashTable is empty
 - 28. getNumberOfItems - return number of items in the hash table
 - 29. add - adds a new node to the hash table
 - 30. remove - remove node with key value
 - 31. clear - remove all entries from the table
 - 32. getItem - returns pointer to the HashNode
 - 33. contains - check if a node with key exists in the table
 - 34. printTable - prints the contents of the hash table to the console and output file

The total number of points for this milestone is 147, which will be based upon the following:

- All file(s) are submitted via Canvas.
- Each submitted/modified file must have student's name (-10% of total milestone points if missing)
- The UML Class Diagram and Sequence Diagram are documented (20)

- Each submitted/modified file must include description of changes made to a program, and its change date (3)
- Program compiles with all of the provided files (1)
- The following methods are documented:
 1. getTable - return the hash table (1)
 2. getList - return the FIFO list (1)
 3. getSize - return the number of items in the CacheManager (1)
 4. isEmpty - Check if the HashTable is empty (1)
 5. add - adds a new node to the CacheManager; if number of items $\geq$ FIFOListSize, then remove the oldest entry (1)
 6. remove - remove node with key value (1)
 7. clear - remove all entries from the CacheManager (1)
 8. getItem - retrieve item from the CacheManager (1)
 9. getMaxCacheSize - retrieve max size of cache from the CacheManager (1)
 10. contains - check if a node with key exists in the table (1)
 11. printCache - prints the current contents of the cache to console and output file (1)
 12. getSize - Get the size of the list (1)
 13. isEmpty - Check if the list is empty (1)
 14. insertAtHead - Adds a new node at the beginning of the list (1)
 15. insertAtTail - Adds a new node at the end of the list (1)
 16. remove - Searches for a node with a specific value and deletes it from the list (1)
 17. removeHeaderNode - Removes header node (1)
 18. removeTailNode - Removes tail node (1)
 19. moveNodeToHead - Moves a specific node to the front (1)
 20. moveNodeToTail - Moves a specific node to the end (1)
 21. clear - Clear the list (delete all nodes) (1)
 22. printList - prints the list from head to tail to the console and output file (1)
 23. reversePrintList - prints the list from tail to head to the console and output file (1)
 24. getTable - return the hash table (1)
 25. getSize - return the size of the hash table (1)
 26. calculateHashCode - perform hashing function (1)
 27. isEmpty - Check if the HashTable is empty (1)
 28. getNumberOfItems - return number of items in the hash table (1)
 29. add - adds a new node to the hash table (1)
 30. remove - remove node with key value (1)
 31. clear - remove all entries from the table (1)
 32. getItem - returns pointer to the HashNode (1)
 33. contains - check if a node with key exists in the table (1)
 34. printTable - prints the contents of the hash table to the console and output file (1)

- The following methods run without errors:
 1. getTable - return the hash table (1)
 2. getList - return the FIFO list (1)
 3. getSize - return the number of items in the CacheManager (1)
 4. isEmpty - Check if the HashTable is empty (1)
 5. add - adds a new node to the CacheManager; if number of items $\geq$ FIFOListSize, then remove the oldest entry (1)
 6. remove - remove node with key value (1)
 7. clear - remove all entries from the CacheManager (1)
 8. getItem - retrieve item from the CacheManager (1)
 9. getMaxCacheSize - retrieve max size of cache from the CacheManager (1)
 10. contains - check if a node with key exists in the table (1)
 11. printCache - prints the current contents of the cache to console and output file (1)
 12. getSize - Get the size of the list (1)
 13. isEmpty - Check if the list is empty (1)
 14. insertAtHead - Adds a new node at the beginning of the list (1)
 15. insertAtTail - Adds a new node at the end of the list (1)
 16. remove - Searches for a node with a specific value and deletes it from the list (1)
 17. removeHeaderNode - Removes header node (1)
 18. removeTailNode - Removes tail node (1)
 19. moveNodeToHead - Moves a specific node to the front (1)
 20. moveNodeToTail - Moves a specific node to the end (1)
 21. clear - Clear the list (delete all nodes) (1)
 22. printList - prints the list from head to tail to the console and output file (1)
 23. reversePrintList - prints the list from tail to head to the console and output file (1)
 24. getTable - return the hash table (1)
 25. getSize - return the size of the hash table (1)
 26. calculateHashCode - perform hashing function (1)
 27. isEmpty - Check if the HashTable is empty (1)
 28. getNumberOfItems - return number of items in the hash table (1)
 29. add - adds a new node to the hash table (1)
 30. remove - remove node with key value (1)
 31. clear - remove all entries from the table (1)
 32. getItem - returns pointer to the HashNode (1)
 33. contains - check if a node with key exists in the table (1)
 34. printTable - prints the contents of the hash table to the console and output file (1)
- The following test cases are processed, and produce expected output (10 per test case; 50 total)

- Extra Credit – use industry standard test program and/or extract test cases, in separate json test file